SONIC bench test: brief results

Setup. Computer → Audient iD14 → Edifier speaker → Earthworks M50 → microphone input 1. Four saved recordings at 96 kHz; approximately 10 cm separation (operator reported). This is an offline analysis.

Timing. All 40 sweeps were accepted. Pooled median application roundtrip: **38.307 ms**; observed range: **38.240–38.375 ms**. The two stream medians differed by 0.125 ms.

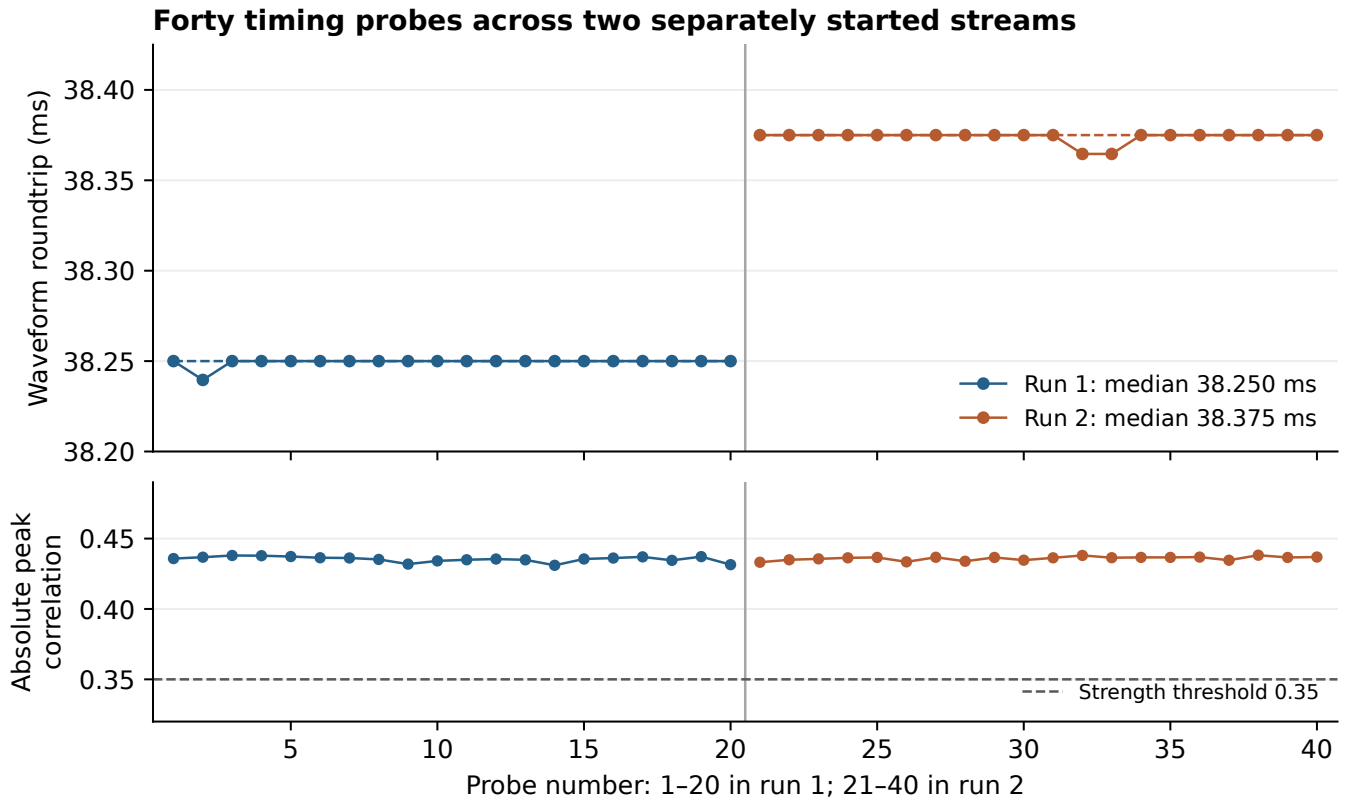


Figure 1: Top: waveform delay estimates for two stream starts. Bottom: matching strength; every probe exceeded the 0.35 threshold. Timing probes were separate 100 Hz–20 kHz sweeps, not SONIC tones.

Meaning. Delay was repeatable over these short runs. It includes computer/interface buffering and the acoustic chain. The observed spread does not establish calibrated sub-millisecond accuracy, isolated speaker delay or neural latency.

Frequency response

SONIC sounds. Two 84-second blocks used a local reconstruction of the paper's 14 Sheep E frequencies (2–27 kHz): 10 ms pips, 2 ms ramps, 16,800 pips total. These are not the original SONIC audio files.

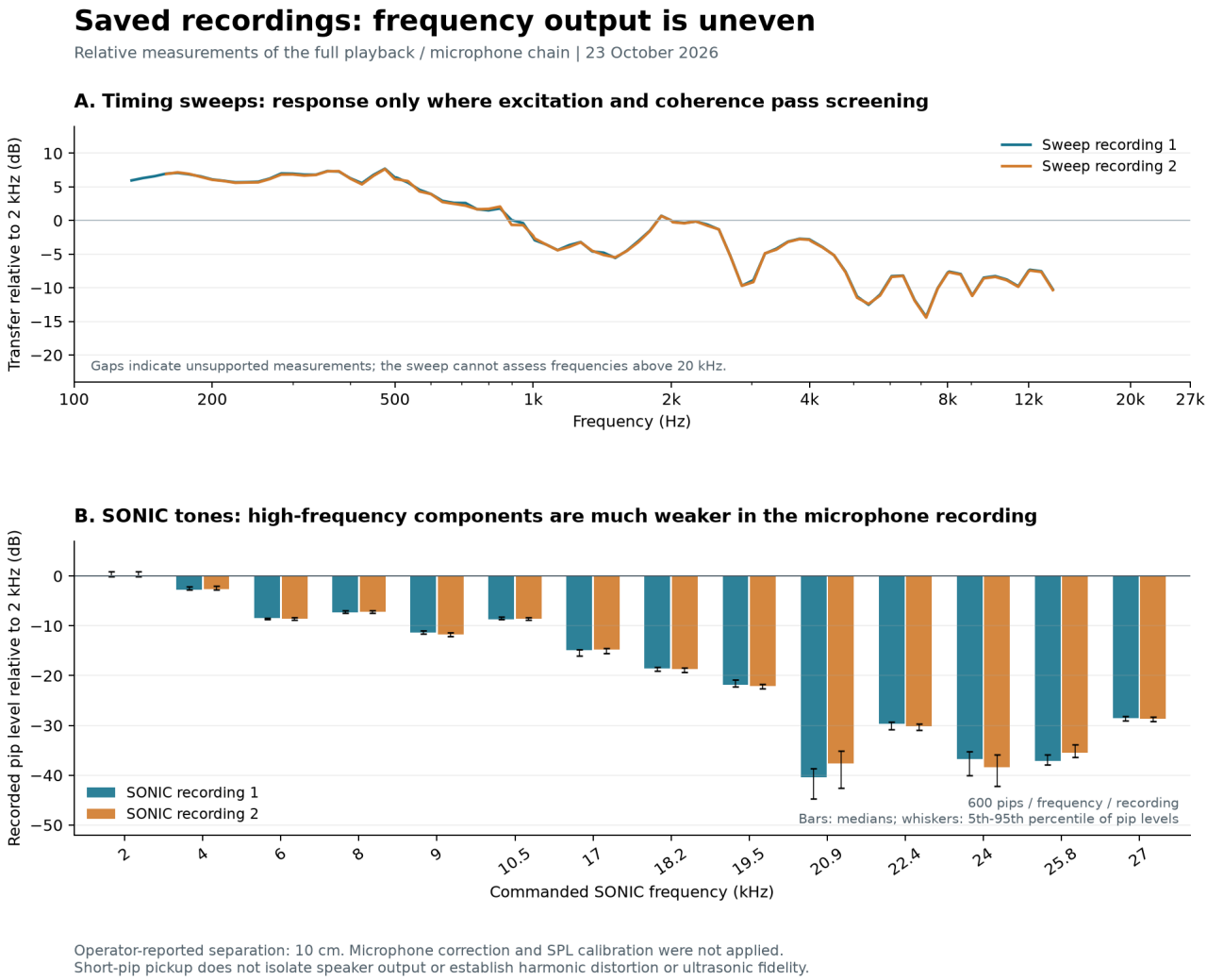


Figure 2: Top: screened sweep-response bands. Bottom: SONIC pip-component medians relative to 2 kHz; whiskers show within-group 5th–95th percentiles, not confidence intervals.

Finding. All 14 commanded components appear in both microphone recordings. Levels are uneven: 17–19.5 kHz was about 15–22 dB below 2 kHz; 20.9–25.8 kHz was 30–40 dB below. These uncorrected full-chain recordings do not establish calibrated SPL, speaker-only response or distortion.

Next. Fix the geometry, calibrate microphone/SPL, test separated tones, and validate timing with an independent electrical reference.

Methods and source records: full saved-data analysis report.